

# Graham Buhrman

PH.D. CANDIDATE · EDUCATIONAL PSYCHOLOGY: QUANTITATIVE METHODS

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## Research Focus

I am a quantitative methodologist working at the intersection of causal inference, measurement, multilevel modeling, and machine learning. My research asks what actually produces the treatment effect heterogeneity that modern methods report and shows that scaling choices, spurious variation, and shrinkage can produce apparent heterogeneity that is difficult to distinguish from genuine substantive effect variation. I develop diagnostics that let applied researchers distinguish among these sources in data they already have. My research bridges causal inference, psychometrics, multilevel modeling, and machine learning and is grounded in the conditions educational effectiveness research presents, including clustered designs, test-scored outcomes, and small sites, where these distortions can disproportionately affect inferences about the students, subgroups, and schools least represented in the data.

## Education

### University of Wisconsin-Madison

PH.D. EDUCATIONAL PSYCHOLOGY

- Dissertation: *Measurement, Regularization, and the Resolution of Causal Claims*
- Advisor: Dr. Jee-Seon Kim

Madison, WI

Anticipated: May 2027

### University of Wisconsin-Madison

MS EDUCATIONAL PSYCHOLOGY

- Advisor: Dr. Jee-Seon Kim

Madison, WI

2022 - 2024

### North Carolina State University

BS INTERDISCIPLINARY STUDIES

- Undergraduate Thesis Advisor: Dr. Vanessa Volpe

Raleigh, NC

2018 - 2022

## Publications

### PEER-REVIEWED JOURNAL ARTICLES

**Buhrman, G. W.**, Tan, H. L., Hanita, M., Campos-Pereira, M., Kennedy, J. L., Zweig, J., Kaiser, A., Waterman, K. P., & Tally, W. (2026). Measuring Historical Empathy in middle and high school students: The Historical Empathy Measure. *Journal of Social Studies Research*, 50(4), 203–215. <https://doi.org/10.1177/0885985X261434163>

Volpe, V. V., Ross, J., Holland, M., Badejoh, O., & **Buhrman, G. W.** (2026). Online health information searching and health advertisement exposure among Black young adult social media users. *Journal of Health Psychology*, 31(5), 2142–2149. <https://doi.org/10.1177/13591053251378224>

Volpe, V. V., **Buhrman, G. W.**, Boaheng, P., Holliday, D., Nick, E. A., & Criss, S. (2023). “Speak[ing] my mind”: Reasons for using Twitter and the online experiences, critical media literacy, and racial identity of Black American emerging adults. *Journal of Media Psychology*, 35(5). <https://doi.org/10.1027/1864-1105/a000387>

Ganguly, I., **Buhrman, G. W.**, Mun, S. K., Kline, E., & Sengupta, S. (2023). Automated error labeling in radiation oncology via statistical natural language processing. *Diagnostics*, 13(7). <https://doi.org/10.3390/diagnostics13071215>

Neukrug, H., Benson, G. P., **Buhrman, G. W.**, & Volpe, V. V. (2022). Affect reactivity and lifetime racial discrimination among Black college students: The role of coping. *American Journal of Orthopsychiatry*, 92(4). <https://doi.org/10.1037/ort0000630>

### PEER-REVIEWED CONFERENCE PROCEEDINGS

**Buhrman, G. W.**, Kim, J.-S., & Lyu, W. (2026). Adaptive shrinkage with Student's t: Limited gains over Gaussian hierarchical linear models. In Kim, J.-S., Wu, H., Sweet, T. M., Wallin, G., Junker, B., Moustaki, I., Harring, J., Bulut, O., Tong, X., & Di Plinio, S. (Eds.), *Proceedings of the International Meeting of the Psychometric Society: The 90th Annual Meeting, Minneapolis, United States, 2025*. <https://doi.org/10.64028/imps2025.W675kxaKDz>

Kim, J.-S., **Buhrman, G. W.**, & Liao, X. (2026). Cluster-specific estimands: Identification and uncertainty in multilevel analysis. In Kim, J.-S., Wu, H., Sweet, T. M., Wallin, G., Junker, B., Moustaki, I., Harring, J., Bulut, O., Tong, X., & Di Plinio, S. (Eds.), *Proceedings of the International Meeting of the Psychometric Society: The 90th Annual Meeting, Minneapolis, United States, 2025*. <https://doi.org/10.64028/imps2025.8C9eDxYWUE>

**Buhrman, G. W.**, Liao, X., & Kim, J.-S. (2025). Using nonparametric regression trees to estimate different forms of heterogeneous treatment effects. In Kim, J.-S., Wu, H., Sweet, T. M., Molenaar, D., Junker, B. W., Moustaki, I., Harring, J., Bulut, O., Tong, X., Wallin, G., & Di Plinio, S. (Eds.), *Proceedings of the International Meeting of the Psychometric Society: The 89th Annual Meeting, Prague, Czech Republic, 2024*. <https://doi.org/10.64028/pulr698375>

Kim, J.-S., **Buhrman, G. W.**, & Liao, X. (2025). Beyond averages: Portraying treatment effect variation. In Kim, J.-S., Wu, H., Sweet, T. M., Molenaar, D., Junker, B. W., Moustaki, I., Harring, J., Bulut, O., Tong, X., Wallin, G., & Di Plinio, S. (Eds.), *Proceedings of the International Meeting of the Psychometric Society: The 89th Annual Meeting, Prague, Czech Republic, 2024*. <https://doi.org/10.64028/xvhr927961>

**Buhrman, G. W.**, Liao, X., & Kim, J.-S. (2024). Exploring conceptual differences between nonparametric estimators of treatment heterogeneity in the context of clustered data. In Wiberg, M., Kim, J.-S., Hwang, H., Wu, H., & Sweet, T. (Eds.), *Quantitative psychology: The 88th Annual Meeting of the Psychometric Society, College Park, Maryland, 2023*. New York: Springer.

## UNDER REVIEW

**Buhrman, G. W.**, & Kim, J.-S. (under review). Identifying exceptional sites in multi-site trials: Boundary conditions and the importance of thresholds and cost ratios for decision-theoretic empirical Bayes. *Journal of Research on Educational Effectiveness*.

**Buhrman, G. W.** (under review). The resolution gap: Global guarantees and local limitations in causal machine learning. *Psychometrika*.

Murphy, D., **Buhrman, G. W.**, & Sedivy, S. (under review). Revisiting undergraduate student reasons for (not) enrolling in language courses. *Second Language Research & Practice*.

Badejoh, O. O., **Buhrman, G. W.**, & Volpe, V. V. (under review). The role of online social support in the association between online racial discrimination and anxiety symptoms in Black adults. *Journal of Media Psychology*.

## Research and Professional Experience

### University of Wisconsin-Madison - Testing & Evaluation Services

Madison, WI

#### PROJECT ASSISTANT

2024-present

- Perform item analyses on Universities of Wisconsin placement tests.
- Describe trends in accommodated testing and utilization of testing space on campus.
- Present analyses and historical data to instructors from the Universities of Wisconsin.

### Development Services Group, Inc.

Bethesda, MD (remote)

#### CONSULTANT

2024-2025

- Wrote and reviewed procedures for calculating and coding effect sizes from crime and justice literature.
- Found and fixed errors in effect size calculation and advised coding decisions.

### Education Development Center

New York City, NY (remote)

#### PROJECT ASSISTANT AND JUNIOR METHODOLOGIST

2023-2025

- Analyzed data from a randomized controlled trial and wrote reports on findings.
- Led the item-response analysis that culminated in a separate publication on the measurement properties of an original instrument.
- Advised team members on statistical analyses and on how to present results.

### University of Wisconsin-Madison - Center for Teaching, Learning, and Mentoring

Madison, WI

#### EVALUATION SPECIALIST

2023-2024

- Evaluated the efficacy of programs for academic staff and faculty via descriptive analysis and data visualization.
- Interviewed academic staff about their experiences with professional development programs.

## Awards, Fellowships, & Grants

|           |                                                                                                                    |                |
|-----------|--------------------------------------------------------------------------------------------------------------------|----------------|
| 2026      | <b>Student Travel Award</b> , University of Wisconsin-Madison School of Education                                  | \$ 1,500       |
| 2024      | <b>Student Travel Award</b> , Psychometric Society                                                                 | \$ 1,250       |
| 2022-2024 | <b>Interdisciplinary Training Program in Education Sciences Fellowship</b> , Institute of Education Sciences       | \$ 36,000/year |
| 2022      | <b>Richard L. Blanton Outstanding Capstone Award</b> , North Carolina State University Honors and Scholars Program |                |
| 2021      | <b>Undergraduate Research Award</b> , North Carolina State University Office of Undergraduate Research             | \$ 2,000       |
| 2021      | <b>Summer Research Award</b> , North Carolina State University Honors and Scholars Program                         | \$ 1,000       |
| 2019-2022 | <b>Caldwell Fellowship</b> , North Carolina State University                                                       | \$ 7,500       |
| 2019-2020 | <b>James F. Kelly Scholarship</b> , North Carolina State University                                                |                |

## Presentations

\* *presenting author(s)*

### INVITED TALKS

2024, October 4. *U.S. history through young people's eyes: a randomized controlled trial of Mission US*. [Seminar presentation] Interdisciplinary Training Program, University of Wisconsin-Madison, Madison, WI.

### CONTRIBUTED PRESENTATIONS

**Buhrman, G. W.\***, & Kim, J.-S. (2026, July 22). *Measurement artifacts in causal machine learning* [Paper presentation]. International Meeting of the Psychometric Society, Seoul, Republic of Korea.

Kim, J.-S.\*, **Buhrman, G. W.**, & Liao, X. (2026, April 11). *Bridging theory, data, and method: A framework for modeling heterogeneous treatment effects in educational research* [Poster presentation]. AERA, Los Angeles, CA, United States.

**Buhrman, G. W.\***, & Kim, J.-S. (2026, April 10). *Toward robust prediction and classification of cluster-specific effects in hierarchical linear modeling* [Paper presentation]. AERA, Los Angeles, CA, United States.

**Buhrman, G. W.\***, Kim, J.-S. & Lyu, W. (2025, July 18). *Can we turn the dial on shrinkage? A comparative study of models for heavy-tailed random effects* [Paper presentation]. International Meeting of the Psychometric Society, Minneapolis, MN, United States.

Kim, J.-S.\*, **Buhrman, G. W.**, Liao, X., & Loh, W. W. (2025, July 18). *A taxonomy of heterogeneity in causal effects* [Paper presentation]. International Meeting of the Psychometric Society, Minneapolis, MN, United States.

Hanita, M., **Buhrman, G. W.\***, Kennedy, J. L.\*, Zweig, J., Tan, H. L., Kaiser, A., & Waterman, K. P. (2024, September 21). *U.S. history through young people's eyes: a randomized controlled trial of Mission US* [Paper presentation]. Society for Research on Educational Effectiveness, Baltimore, MD, United States.

**Buhrman, G. W.\***, Liao, X., & Kim, J.-S. (2024, July 18). *Monotonicity matters: Evaluating heterogeneous treatment effect estimation via nonparametric regression tree methods* [Paper presentation]. International Meeting of the Psychometric Society, Prague, Czech Republic.

Kim, J.-S.\*, Liao, X., **Buhrman, G. W.**, & Loh, W. W. (2024, July 18). *An overarching framework for examining diverse forms of heterogeneous causal effects* [Paper presentation]. International Meeting of the Psychometric Society, Prague, Czech Republic.

**Buhrman, G. W.\***, Liao, X., & Kim, J.-S. (2023, July 25). *Exploring conceptual differences between nonparametric estimators of treatment heterogeneity in the context of clustered data* [Poster session]. International Meeting of the Psychometric Society, College Park, MD, United States.

**Buhrman, G. W.\***, & Volpe, V. V. (2022). Stress from hateful anti-Black Twitter content and the mental health of Black young adult Twitter users. [Poster presentation]. North Carolina State University, Interdisciplinary Studies Symposium, Raleigh, NC.

## Graduate Teaching Experience \_\_\_\_\_

|             |                                                                                   |                    |
|-------------|-----------------------------------------------------------------------------------|--------------------|
| Fall 2026   | <b>Introduction to the Design of Educational Experiments</b> , Teaching Assistant | <i>Madison, WI</i> |
| Fall 2026   | <b>Regression Models in Education</b> , Teaching Assistant                        | <i>Madison, WI</i> |
| Spring 2025 | <b>Hierarchical Linear Modeling</b> , Teaching Assistant                          | <i>Madison, WI</i> |
| Spring 2025 | <b>Field Experiments in Education Research</b> , Teaching Assistant               | <i>Madison, WI</i> |

## Outreach & Professional Development \_\_\_\_\_

### SERVICE AND OUTREACH

|           |                                                           |                           |
|-----------|-----------------------------------------------------------|---------------------------|
| 2023-2025 | <b>Undergraduate Research Symposium</b> , Volunteer       | <i>Madison, WI</i>        |
| 2020-2022 | <b>Campus Conversations Project</b> , Student Facilitator | <i>Raleigh, NC</i>        |
| 2019      | <b>Neighbor2Neighbor</b> , Enrichment Mentor              | <i>Raleigh, NC</i>        |
| 2019      | <b>Fundación Comunitaria del Bajío</b> , Volunteer        | <i>Guanajuato, Mexico</i> |

### AD HOC REVIEWER

Psychometrika  
Advances in Methods and Practices in Psychological Science

### PROFESSIONAL MEMBERSHIPS

American Statistical Association  
Psychometric Society  
Society for Research on Educational Effectiveness  
American Educational Research Association

## References \_\_\_\_\_

### **Jee-Seon Kim**

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### **Daniel Bolt**

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### **James Wollack**

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